SATWIK KUKADOLLI

Backend & Full-Stack Software Engineer

Dharwad, Karnataka, India

+91-6361559598 | satwikkukadolli@gmail.com | GitHub: github.com/Satwik0909 | LinkedIn: linkedin.com/in/satwikkukadolli

PROFESSIONAL SUMMARY

Backend Software Engineer with 1.5+ years of experience building production SaaS platforms, REST APIs, and multi-tenant systems. Skilled in Python (Django/DRF), Node.js, PostgreSQL, and MongoDB. Experienced in building RBAC systems, payment integrations, CI/CD pipelines, and performance improvements. Uses modern developer tools, including AI-assisted scaffolding, to ship features faster without lowering quality.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), Java, HTML5, CSS3
Backend: Django, Django REST Framework (DRF), Node.js, Express.js
Frontend: React.js, Tailwind CSS, Bootstrap
Databases: PostgreSQL, MongoDB (Atlas), MySQL — indexing, query optimisation, aggregation pipelines
Auth & Security: JWT, OAuth2, RBAC, Session Management, API security hardening, multi-tenant isolation
DevOps & Tools: Git, GitHub Actions (CI/CD), Nginx, VPS, Postman, Render, Netlify
Core Concepts: REST APIs, Multi-Tenant Architecture, System Design, Performance Optimisation, Agile/Scrum

PROFESSIONAL EXPERIENCE

Sajre Edutech Pvt. Ltd.
Software Engineer – Backend Sep 2025 – March 2026

- **Multi-tenant SaaS backend** for school institutions across Karnataka, using **schema-per-tenant PostgreSQL** to keep each tenant's data separate.
- Built and maintained **30+ REST APIs** using **Django REST Framework**; used AI tools to cut boilerplate setup time, reducing feature delivery time by ~40%.
- Built a multi-role **RBAC system** to keep tenant data separate; wrote permission-boundary test cases checked on every deployment.
- Built an automated **bulk PDF certificate pipeline** that delivered outputs within a 2-hour SLA; template improvements cut manual QA cycles by 60%.
- Set up **Razorpay payment integration** with idempotent transaction handling and real-time checks to flag unusual payment activity.
- Set up **WhatsApp Business API** for real-time notifications and referral tracking; message personalisation improved click-through by ~22%.
- Set up **Nginx reverse proxy** with VPS hosting and **GitHub Actions CI/CD** pipelines; kept 99%+ uptime and added runbooks to speed up recovery.

Ironhawks Technologies
Software Engineer – Full-Stack Apr 2025 – Aug 2025

- Built full-stack applications using **Node.js (Express.js)** and **React.js**; used AI scaffolding tools to build CRUD modules faster while keeping test coverage.
- Built secure REST APIs with **JWT authentication** and role-based authorisation; added PR review tools to flag OWASP Top 10 issues before merge.
- Improved MongoDB query plans using composite indexes and explain analysis, improving average API response time by ~25% under production load.
- Built reusable third-party API service adapters and generated **OpenAPI documentation**, cutting new-engineer onboarding time by ~30%.
- Improved React.js frontend performance through lazy loading and code-splitting, cutting initial load times by ~30% and reaching Lighthouse scores above 90.

Letzstepin Pvt. Ltd.
Software Engineer – Full-Stack Nov 2024 – Mar 2025

- Built responsive **React.js** UI components for a VR-based learning platform; used AI tools to cut screen prototyping time from days to under 2 hours per feature.
- Cut repeated UI components by 35% through a reusable-component refactor.
- Improved cross-browser compatibility and **WCAG accessibility**; found and fixed 15+ accessibility issues through structured audits.
- Took part in Agile sprints, including code reviews and sprint planning; introduced PR summary templates that cut review turnaround time by ~20%.

TECHNICAL PROJECTS

School ERP SaaS Platform

Personal Project · Django · DRF · PostgreSQL · React.js · GitHub Actions

- Built a multi-tenant ERP using **schema-per-tenant PostgreSQL**, giving each school fully separate data with no cross-tenant leakage.
- Set up composite B-tree indexes on high-traffic tables, reaching **sub-100ms API responses** under 500+ simulated concurrent requests in load testing.
- Built role-based dashboards for Super Admin, School Admin, Teacher, and Student roles; wrote **RBAC test matrices** covering 200+ permission combinations.
- Set up **GitHub Actions CI/CD** pipelines with lint, test, and deploy checks; reached 80%+ test coverage from Sprint 1.

Fitness Membership Management System

Personal Project · MongoDB · Node.js · Express.js · React.js · JWT

- Built a full-stack **MERN application** with JWT auth, role-based dashboards, attendance tracking, and automated billing logic.
- Built **MongoDB aggregation pipelines** for revenue analytics; pipeline and index improvements cut dashboard query time by 35%, enabling real-time chart updates.
- Built a subscription lifecycle system with atomic transactions handling plan upgrades, downgrades, cancellations, and prorated billing — removing inconsistent membership states.
- Set up automated WhatsApp and email reminders 3 days before membership expiry to reduce lapsed renewals.

Banquet Hall Booking System

Personal Project · MongoDB · Node.js · Express.js · React.js · JWT

- Built a real-time booking platform with **atomic MongoDB transactions** that removed double-booking issues during concurrent reservation requests.
- Built dynamic pricing logic that adjusts hall rates by date, time slot, and add-on services, enabling upsell opportunities at checkout and increasing average booking value.
- Built REST APIs with input validation, structured error codes, and rate limiting; generated **OpenAPI documentation** that cut third-party integration effort by 2 full sprints.
- Wrote concurrency edge-case tests covering scenarios often missed in manual QA, including simultaneous bookings for the same slot across multiple sessions.

Service Request Portal — 12-hr Hackathon

HackFusion 2025 Finalist · Django · DRF · JWT · PostgreSQL · React.js

- Built and delivered a JWT-secured service request portal with admin dashboards, SLA breach tracking, and automated escalation workflows within a 12-hour hackathon window.
- Set up 8 core REST API endpoints in under 90 minutes, freeing time to build 3 bonus features beyond the original MVP scope.
- Placed in the top cohort among 50+ competing teams; judges recognised production-ready code quality within the time constraint.

EDUCATION

IBMR College, Hubli, Karnataka 2022 – 2025

Bachelor of Computer Applications (BCA) – CGPA: 7.0